Worksheet 2-8 & 2-9 B Product and Quotient Rules #2
AP Calculus Name_____

x	-4	-1	1	7
$g(x)$	1	2	-3	0
$g'(x)$	5	-4	2	-1
$g''(x)$	2	6	3	-5

Selected values of the twice differentiable function $g(x)$ are given in the table above along with $g'(x)$ and $g''(x)$. Use this information to find the following.

1. $h(x) = x^2 g(x)$. Find $h'(-4)$.

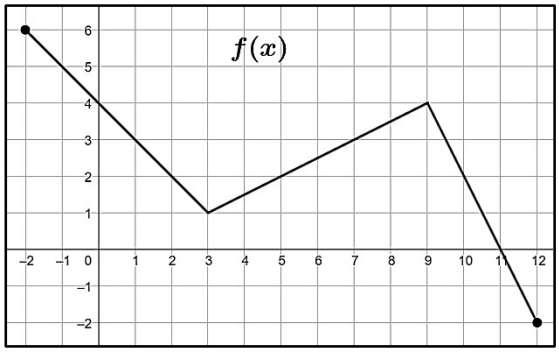
2. $k(x) = 2\cos(x)g'(x)$. Find $k'(7)$.

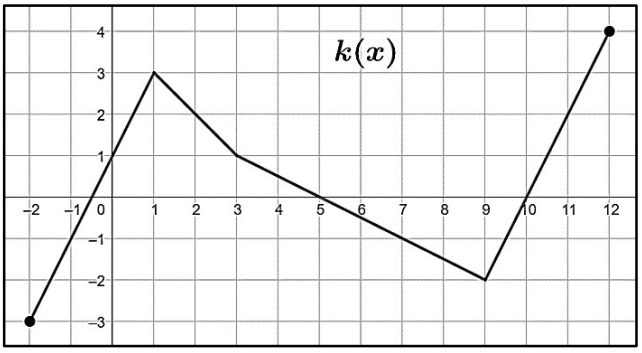
3. $f(x) = x \cdot \ln(x) - g(x)$. Find $f'(1)$.

4. $p(x) = g(x)g'(x)$. Find $p'(-1)$.

5. $m(x) = \dfrac{g(x)}{x+1}$. Find $m'(1)$.

6. $r(x) = \dfrac{g'(x)}{g(x)}$. Find $r'(-4)$.





The graphs of $f(x)$ and $k(x)$ are shown above and consist of line segments. Let $g(x) = f(x)k(x)$.

7. Find $g'(7)$.

8. Find $g'(1)$.

9. Let $h(x) = x^2 f(x)$. Find $h'(10)$.

10. Let $m(x) = k(x)e^x$. Find $m'(-1)$.

11. Let $r(x) = -2x^3 f(x)$. Find the instantaneous rate of change of $r(x)$ at $x = 5$.

12. Let $s(x) = x^2 - f(x)$. Find the average rate of change of $s(x)$ over the interval $[2, 7]$.

13. Let $p(x) = -3\cos(x)\, k(x) + 4x$. Write an equation of the line tangent to p of at $x = 0$.